

NecessityEntrepreneurs: Paper Math Referee Findings

Draft slides for coauthor discussion

Source of truth: Chivers et al. (2025) Necessity Entrepreneurship.lyx

February 18, 2026

Objective and Scope

- ▶ Goal: identify remaining **paper-math** issues in the current .lyx draft.
- ▶ Scope: equations/definitions only (not code implementation details).
- ▶ Baseline checked: main text and appendix consistency.
- ▶ Output: numbered issues with exact equation labels and line locations.

Good News: Major Old Errors Look Fixed

- ▶ Firm FOC no longer has stray $-rk$ term.
- ▶ Main closed forms now coherent:
 - ▶ labor demand at FOC (lyx:1147)
 - ▶ profit at profitj (lyx:1175)
- ▶ Unified budget sign now uses $(1 + r - \delta)a$ in eq:c_worker-1 (lyx:1469).

Issue 1 (High): Main vs Appendix Technology Mismatch

- ▶ Main text uses production proportional to $xk^\alpha(zn)^\gamma$:
 - ▶ `maxP (lyx:1078)`, `profitj (lyx:1175)`
- ▶ Appendix uses $x^{1-\alpha-\gamma}k^\alpha(zn)^\gamma$:
 - ▶ `eq:xstar_u`, `eq:xstar_n (lyx:8866–8867)`
 - ▶ `eq:profit_entrepreneur (lyx:9111)`
- ▶ This can shift talent returns, cutoffs, and entrepreneurship shares.

Issue 2 (High): Entrepreneur Continuation State Inconsistency

- ▶ State labels define entrepreneur as a distinct state:
 - ▶ main text (lyx:1373–1382), appendix (lyx:8786–8795)
- ▶ But Bellman terms send entrepreneur choice to $i'_w = 0$:
 - ▶ main Bellman (lyx:1454–1457)
 - ▶ appendix Bellman eq: V_{overall} block (lyx:8907–8919)
- ▶ Need one clear transition convention for entrepreneurship status.

Issue 3 (Medium): Formal Typo in Appendix Bellman

- ▶ In eq:V_iw0_xh, continuation state is written as $i'_w = w$.
- ▶ Location: lyx:9094.
- ▶ This is almost surely a typo, but should be fixed for formal validity.

Issue 4 (High): Cutoff in Flow Terms vs Dynamic Values

- ▶ Appendix cutoff equations set $\tilde{w} = \pi(x^*)$:
 - ▶ eq:xstar_u, eq:xstar_n (lyx:8866–8867)
- ▶ Dynamic decision is defined by Bellman value comparison:
 - ▶ eq:V_overall block (lyx:8907–8922)
- ▶ Need either:
 - ▶ value-equality cutoffs, or
 - ▶ explicit assumptions that justify flow-equality shortcut.

Issue 5-6 (Medium): Notation Consistency Gaps

- ▶ Tax symbol inconsistency:
 - ▶ uses τ_y in worker budgets (lyx:1259, 1299)
 - ▶ uses τ in expected worker income (lyx:1496–1500, 8885–8886)
- ▶ Bellman continuation notation drops state components:
 - ▶ LHS: $V(a, h, z, x, i_w)$
 - ▶ continuation written as $V(a', x', i'_w)$ at lyx:1454–1457
- ▶ Likely fixable as notation cleanup, but important for replication clarity.

Could These Issues Change Results?

- ▶ **Most likely to matter quantitatively:**
 - ▶ Issue 1 (technology mismatch)
 - ▶ Issue 2 (state transition inconsistency)
 - ▶ Issue 4 (flow vs value cutoff logic)
- ▶ **Mostly clarity/formal correctness:**
 - ▶ Issue 3 (typo $i'_w = w$)
 - ▶ Issue 5-6 (notation consistency)
- ▶ If implementation follows corrected logic, some may be documentation-only.

Numbered Edit Plan in the Paper

1. Unify production/profit form across main text and appendix.
2. Fix entrepreneur continuation state mapping in both Bellman blocks.
3. Correct eq: V_{iw0_xh} typo ($i'_w = w$).
4. Rewrite cutoff derivation in value-function terms (or state assumptions).
5. Standardize tax symbol usage (τ vs τ_y).
6. Harmonize continuation notation to keep full state accounting explicit.

Deliverables in Referee Folder

- ▶ `PaperMathReferee_2026-02-18/PAPER_MATH_REFEREE_REPORT.tex`
- ▶ `PaperMathReferee_2026-02-18/PAPER_MATH_REFEREE_REPORT.pdf`
- ▶ `CodeAudit_2026-02-18/CODE_REFEREE_REPORT.md`
- ▶ `CodeAudit_2026-02-18/PAPER_CODE_CROSSWALK.csv`
- ▶ `CodeAudit_2026-02-18/FIX_CHECKLIST.md`

Next step: resolve Issues 1/2/4 first, then rerun baseline tables.